

Kamlesh Kumar

Full-Stack Software Engineer



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SUMMARY

Software engineer with a solid computer science foundation in data structures and algorithms, system design, and distributed systems. I have worked across the stack, from C code in PostgreSQL's high-availability tooling to production products in TypeScript, Next.js, and Python. In 2.5 years of professional work, I built the Google Pay integration end to end for a payment gateway processing tens of millions of euros, and helped cut its payment processing time by 11%.

TECHNICAL SKILLS

Languages	TypeScript, Python, JavaScript, Go, C/C++
Frontend	React, Next.js, Tailwind CSS
Backend	Node.js, Next.js Server Actions, FastAPI, Prisma, REST API design, OAuth 2.0, JWT, webhooks, CRON jobs, Redis caching
Databases	PostgreSQL, ORMs, MongoDB, MSSQL, YugabyteDB; schema design, indexing, query optimization
Cloud & DevOps	AWS (CDK, ECS Fargate, S3, CloudFront), Terraform, Docker, GitHub Actions, Jenkins, CI/CD pipelines
AI	LLM agents, Model Context Protocol (MCP) tool-calling, LangChain, PyTorch, Hugging Face, fine-tuning; Claude Code, Cursor

WORK EXPERIENCE

Full-Stack Developer (HR-Tech)

Espoo, Finland
Mar 2026 – Jun 2026

A startup recruitment platform connecting talent with companies. Joined as the only engineer and a **core team member**, owning the platform end-to-end while it handled real user traffic.

- Built a **multi-user messaging system** between companies and talent, featuring in-chat acceptance/rejection workflows and scheduled Inngest background jobs to deliver **rate-limited email notifications**.
- Implemented a self-serve applicant workflow** so companies can independently manage hiring on the platform, with excel data export and bulk in-platform messaging announcements.
- Implemented **Upstash Redis caching layer** and Sentry error tracking from scratch, boosting **system performance** and **observability**.
- Traced a race condition** between two signup paths that caused mismatched user IDs across database tables and metadata, diagnosed the root cause, and shipped a permanent fix.
- Debugged and resolved **pre-existing inconsistent data across the auth and application layers**.
- Participated in product and market discussions, shaped decisions, built a full understanding of the business alongside the code, and hired two new developer interns.

Tech stack: Next.js, TypeScript, WebSockets, Supabase, Prisma, nginx, GitHub Actions, Docker Swarm, Terraform

Full Stack Engineer (Fin-Tech)

Ireland, Remote
Jun 2024 – Jul 2025

PCI DSS Level 1 payment gateway, 25+ years old, processing tens of millions of euros in 20+ currencies. Worked with teams in Ireland, Cyprus, and Pakistan.

- Cut end-to-end payment processing time by **11%**: optimized slow SQL queries and indexes, made log writes and transaction status updates asynchronous so payments no longer waited on them, and removed legacy VBScript.
- Built the core gateway's **Google Pay** integration end to end, from the payment API and token handling to the transaction flow and webhooks, and worked with Google's technical teams to qualify it as a participating processor.
- Helped complete the **Apple Pay integration** in the core gateway.
- Designed and built **PCI-compliant REST APIs** secured with JWT, so the mobile app could connect directly to the core gateway. This improved data consistency and unified the experience across platforms.
- Worked directly with gateway customers** on their white-label setups, each isolated in its own database: diagnosed notification, webhook, risk rule, and performance issues, and shipped the fixes to production.

Tech stack: TypeScript, React, Python, Java, Spring Boot, MSSQL, YugabyteDB (database-per-tenant), Azure, Docker

Software Engineer Intern, Open Source

Bitnine Global Inc, Part-time

South Korea, Remote

Nov 2022 – Nov 2023

Contributed [11 merged pull requests](#) to **Apache AGE**, an open-source graph database extension for PostgreSQL.

- Worked on high availability for AGE through Pgpool-II (load balancing, read/write splitting): implemented Cypher function extraction and added regression tests for Cypher load balancing, in C.
- Wrote REST APIs, routing, and sessions in Go for AgeViewer-Go, the desktop app for managing AGE databases: connecting and disconnecting databases, reading graph metadata, and returning query results.
- Built AgeDB-Cloud, a web app for using AGE in the browser without installing it, on a DigitalOcean VM. [🔗](#)

Tech stack: C, Go, PostgreSQL, Pgpool-II, React, Node.js, MongoDB, PM2

EDUCATION

Master of Science in Computer Science

Sep 2025 – Mar 2027 (expected)

University of Helsinki (GPA: 4.9/5)

Helsinki, Finland

- **Thesis with Optomed Oy:** deep learning for detecting hypertensive retinopathy in cats from retinal fundus images, funded by an **AI thesis grant** from the Technology Industries of Finland Centennial Foundation.
- 100% merit scholarship · 85 of 90 course credits completed · [Transcript](#)
- **Courses:** Software Architecture, Full-Stack Development, Distributed Systems, Engineering of Machine Learning.

Bachelor of Science in Computer Science

Sep 2020 – Jun 2024

National University of Sciences and Technology (NUST)

Islamabad, Pakistan

- 100% merit scholarship · [Transcript](#)

PROJECTS

- **Cloud Deployment as Code on AWS** [🔗](#): Containerized web app on AWS as code: a VPC, ECS Fargate behind a load balancer, and a React frontend on S3 and CloudFront, all brought up by one command.
Tools: AWS CDK, TypeScript, Docker, ECS Fargate, S3, CloudFront
- **Nodetalk (Distributed Chat Systems)**: Three-node prototype where the same Python server runs on three university machines and each node connects to the other two over WebSockets. A chat message sent by a client to any node is replicated to all three, so every node holds every message in memory. [GitHub](#)
Tools: Python, WebSockets
- **AI Agent with MCP Tool-Calling** [🔗](#): Voice and text agent that acts on a user's Google Calendar and personal context through custom MCP tools, running in a Telegram Mini App and on the web.
Tools: TypeScript, Next.js, MongoDB, MCP, OpenAI Realtime API
- **Compiler for a Custom Language** [🔗](#): Compiler in Python from a custom language to x86-64 assembly, every stage written by hand from tokenizer to code generation, packaged in Docker with a full test suite.
Tools: Python, Docker, x86-64 assembly

CERTIFICATIONS & LANGUAGES

- **Certifications:** DevOps with Docker (2026) [🔗](#) and DevOps with Kubernetes (Ongoing), University of Helsinki; PostgreSQL Indexes, Percona (2023) [🔗](#)
- **Languages:** English (full professional, IELTS 7.0) · Finnish (basic, improving)

LEADERSHIP & ACTIVITIES

- **United Nations Millennium Fellowship, Campus Director (2023)**: Led 17 fellows across 5 projects, after selection into the top 1% of applicants. [🔗](#)
- **Pathways Pre-Incubator Programme (Helsinki Incubators)**: Completed the entrepreneurship programme. [🔗](#)
- **Slush Volunteer (2025)**: Supported Founder Days at Slush. [🔗](#)